Appendix A. Model Definitions and Theoretical Foundation

This appendix provides the theoretical background, mathematical equations, notation descriptions and remarks for the four forecasting models used in this study.

A.1 Linear Regression

Linear Regression is one of the most widely used statistical forecasting models. The model assumes that a linear relationship exists between the dependent variable and one or more independent variables. In financial forecasting, it serves as an important benchmark model because of its simplicity and interpretability.

Mathematical Equation

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \varepsilon$$

Description of Symbols

Symbol	Description
Y	Predicted stock price or forecast value
β_0	Intercept (constant term)
β_i	Regression coefficient
X_i	Independent variable / predictor
n	Number of predictors
ε	Random error term

Remark

Linear Regression assumes a linear relationship between predictors and the target variable. The model is easy to interpret but may struggle to capture complex non-linear market dynamics.

A.2 ARIMA (AutoRegressive Integrated Moving Average)

ARIMA is a classical time-series forecasting model that combines autoregressive, differencing and moving-average components. ARIMA models are commonly used in financial forecasting to analyse historical price movements and identify temporal patterns.

Mathematical Equation

$$Y_t = c + \Sigma(\varphi_i Y_{t-i}) + \Sigma(\theta_j \varepsilon_{t-j}) + \varepsilon_t$$

Description of Symbols

Symbol	Description
Y_t	Current observation at time t
Y_{t-i}	Lagged observation
c	Constant term
p	Autoregressive order
d	Differencing order
q	Moving-average order
φ_i	Autoregressive coefficient
θ_j	Moving-average coefficient
ε_t	Current error term
ε_{t-j}	Previous error term

Remark

ARIMA(p,d,q) consists of:
p = autoregressive order
d = differencing order
q = moving-average order

For example, ARIMA(2,1,2) indicates two autoregressive terms, one differencing operation, and two moving-average terms.

A.3 Random Forest

Random Forest is an ensemble machine-learning algorithm that combines a large number of decision trees and aggregates their forecasts. The method is particularly effective for capturing complex non-linear relationships in financial markets.

Mathematical Equation

$$\hat{Y} = (1/B) \times \Sigma T_{\beta}(X)$$

Description of Symbols

Symbol	Description
$\hat{Y}$	Final predicted value
B	Number of decision trees
$T_{\beta}(X)$	Prediction from tree b
X	Input feature vector
b	Tree index

Remark

Random Forest reduces forecasting variance by averaging predictions from multiple decision trees. The model often performs well on complex financial datasets with non-linear relationships.

A.4 Long Short-Term Memory (LSTM)

LSTM is a deep-learning architecture belonging to the recurrent neural network (RNN) family. LSTM introduces memory cells and gating mechanisms that allow the network to learn long-term temporal dependencies within time-series data.

Mathematical Equations

Forget Gate:
$f_t = \sigma(Wf[h_{t-1},x_t] + bf)$

Input Gate:
$i_t = \sigma(Wi[h_{t-1},x_t] + bi)$

Cell State Update:
$C_t = f_t C_{t-1} + i_t \tilde{C}_t$

Output Gate:
$o_t = \sigma(Wo[h_{t-1},x_t] + bo)$

Hidden State:
$h_t = o_t \tanh(C_t)$

Description of Symbols

Symbol	Description
x_t	Input vector at time t
h_t	Hidden state
h_{t-1}	Previous hidden state
C_t	Cell memory state
C_{t-1}	Previous cell state
f_t	Forget gate
i_t	Input gate
o_t	Output gate
$\tilde{C}_t$	Candidate cell state
Wf	Forget gate weight matrix
Wi	Input gate weight matrix
Wo	Output gate weight matrix
bf	Forget gate bias
bi	Input gate bias
bo	Output gate bias
σ	Sigmoid activation function
tanh	Hyperbolic tangent activation function

Remark

LSTM overcomes the vanishing-gradient problem encountered by traditional RNNs and is highly effective for modelling long-term dependencies in financial time-series forecasting.

Appendix A Summary Comparison

Model	Category	Key Strength	Key Limitation
Linear Regression	Statistical	High Interpretability	Captures only linear relationships
ARIMA	Time-Series	Trend Modelling	Requires stationary data
Random Forest	Machine Learning	Captures non-linear relationships	Less interpretable
LSTM	Deep Learning	Handles long-term dependencies	Computationally expensive